Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Force pair

Meaning in my words How would you explain force pair to a friend?	**How can you tell?** Earth pulls an apple down. What is the partner force, and which object feels it?
Example or sketch How would you label the pair when a hand pushes a cart?	**Non-example and why** Are the table's push and Earth's pull on one book a third-law pair?

Receivers

Meaning in my words In an interaction pair, what are the two receivers?	**How can you tell?** In "the wall pushes on my hand," where does that force arrow belong?
Example or sketch When my hand pushes on the wall, which object receives that other force?	**Non-example and why** Can I draw both contact forces of that pair on the hand?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Mass effect

Meaning in my words	How can you tell?
How does a cart's mass affect its response to a push?	Two skaters push off with equal net force magnitudes. Which skater has greater acceleration?
Example or sketch	**Non-example and why**
A 10 N net force acts separately on 2 kg and 5 kg carts. Compare their accelerations.	The lighter cart accelerates more. Does that prove it felt the larger interaction force?

System

Meaning in my words	How can you tell?
How would you explain system to a friend?	Draw one boundary around two interacting carts. Where are their bumper forces relative to it?
Example or sketch	**Non-example and why**
If a hand pushes one cart from outside that boundary, where does its force go?	When you choose just one cart, can you still drop the force from the other cart?

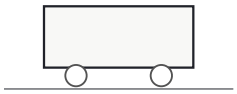
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER

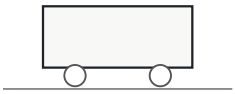
Name _____

Label both receivers first. Draw equal force arrows on different objects, pointing toward one another for gravity.

Receiver A



Receiver B



My marks mean

Two pulls. Two receivers.

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Place an interaction pair on its two receivers.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

Put each gravitational arrow on the body that receives it. Send the animator two corrected object panels.

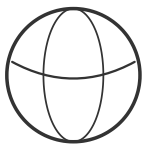
SCIENCE GRAPHICS PRODUCER ☐

Write a split-screen caption naming who pulls whom. Explain why equal force does not require equal acceleration.

Point to a clue. Keep your first marks; add the update using the color key.

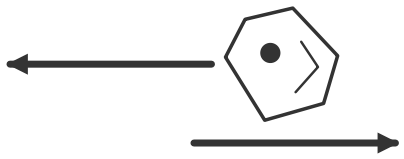
Earth pulls on XJ. A public animation shows both gravitational arrows on XJ, with the caption “they cancel.”

EARTH PANEL



Earth

XJ PANEL



PROPOSAL: “They cancel.”

01 OUR FIRST CALL

Do the two arrows cancel on XJ? Explain using your corrected panels.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Which arrow belongs in the updated shot? Name its source and receiver.

Switch speakers. If Earth and XJ are one combined system, which forces in this interaction are internal to that system?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks